

Also as expected, the error for given method decreases as T decreases. However, perhaps the most noteworthy feature of Table 1.1 is *how* the integration error decreases with T . We can see that with rectangular integration, when T is halved, the integration error is also halved. With trapezoidal integration, when T is halved, the integration error decreases by a factor of four. With Runge–Kutta integration, when T is halved, the integration error decreases by a factor of 16. We conclude that (in general) the error of rectangular integration is proportional to T , the error of trapezoidal integration is proportional to T^2 , and the error of Runge–Kutta integration is proportional to T^4 .

Table 1.1 Example 1.5 results. Percent errors when numerically integrating $\dot{x} = \cos t$ from $t = 0$ to $t = 1$, for various integration algorithms, and for various time step sizes T .

	$T = 0.1$	$T = 0.05$	$T = 0.025$
Rectangular	2.6	1.3	0.68
Trapezoidal	0.083	0.021	0.0052
Fourth-order Runge–Kutta	3.5×10^{-6}	2.2×10^{-7}	1.4×10^{-8}

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1.6 STABILITY

In this section, we review the concept of stability for linear time-invariant systems. We first deal with continuous-time systems in Section 1.6.1, and then discrete-time systems in Section 1.6.2. We state the important results here without proof. The interested reader can refer to standard books on linear systems for more details and additional results [Kai80, Bay99, Che99].

1.6.1 Continuous-time systems

Consider the zero-input, linear, continuous-time system

$$\begin{aligned}\dot{x} &= Ax \\ y &= Cx\end{aligned}\tag{1.118}$$

The definitions of marginal stability and asymptotic stability are as follows.

Definition 1 *A linear continuous-time, time-invariant system is marginally stable if the state $x(t)$ is bounded for all t and for all bounded initial states $x(0)$.*

Marginal stability is also called Lyapunov stability.

Definition 2 *A linear continuous-time, time-invariant system is asymptotically stable if, for all bounded initial states $x(0)$,*

$$\lim_{t \rightarrow \infty} x(t) = 0\tag{1.119}$$

The above two definitions show that a system is marginally stable if it is asymptotically stable. That is, asymptotic stability is a subset of marginal stability. Marginal stability and asymptotic stability are types of internal stability. This is because they deal with only the state of the system (i.e., the internal condition of the system) and do not consider the output of the system. More specific categories of internal stability (e.g., uniform stability and exponential stability) are given in some books on linear systems.

Since the solution of Equation (1.118) is given as

$$x(t) = \exp(At)x(0) \quad (1.120)$$

we can state the following theorem.

Theorem 1 *A linear continuous-time, time-invariant system is marginally stable if and only if*

$$\lim_{t \rightarrow \infty} \exp(At) \leq M < \infty \quad (1.121)$$

for some constant matrix M . This is just a way of saying that the matrix exponential does not increase without bound.

The “less than or equal to” relation in the above theorem raises some questions, because the quantities on either side of this mathematical symbol are matrices. What does it mean for a matrix to be less than another matrix? It can be interpreted several ways. For example, to say that $A < B$ is usually interpreted to mean that $(B - A)$ is positive definite.⁵ In the above theorem we can use any reasonable definition for the matrix inequality and the theorem still holds.

A similar theorem can be stated by combining Definition (2) with Equation (1.120).

Theorem 2 *A linear continuous-time, time-invariant system is asymptotically stable if and only if*

$$\lim_{t \rightarrow \infty} \exp(At) = 0 \quad (1.122)$$

Now recall that $\exp(At) = Q \exp(\hat{A}t) Q^{-1}$, where Q is a constant matrix containing the eigenvectors of A , and $\hat{A}$ is the Jordan form of A . The exponential $\exp(\hat{A}t)$ therefore contains terms like $\exp(\lambda_i t)$, $t \exp(\lambda_i t)$, $t^2 \exp(\lambda_i t)$, and so on, where λ_i is an eigenvalue of A . The boundedness of $\exp(At)$ is therefore related to the eigenvalues of A as stated by the following theorems.

Theorem 3 *A linear continuous-time, time-invariant system is marginally stable if and only if one of the following conditions holds.*

1. *All of the eigenvalues of A have negative real parts.*
2. *All of the eigenvalues of A have negative or zero real parts, and those with real parts equal to zero have a geometric multiplicity equal to their algebraic multiplicity. That is, the Jordan blocks that are associated with the eigenvalues that have real parts equal to zero are first order.*

Theorem 4 *A linear continuous-time, time-invariant system is asymptotically stable if and only if all of the eigenvalues of A have negative real parts.*

⁵Sometimes the statement $A < B$ means that every element of A is less than the corresponding element of B . However, we will not use that definition in this book.

■ **EXAMPLE 1.6**

Consider the system

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} x \quad (1.123)$$

Since the A matrix is upper triangular, we know that its eigenvalues are on the diagonal; that is, the eigenvalues of A are equal to 0, 0, and -1 . We see that the system is asymptotically unstable since some of the eigenvalues are nonnegative. We also note that the A matrix is already in Jordan form, and we see that the Jordan block corresponding to the 0 eigenvalue is second order. Therefore, the system is also marginally unstable. The solution of this system is

$$\begin{aligned} x(t) &= \exp(At)x(0) \\ &= \begin{bmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} x(0) \end{aligned} \quad (1.124)$$

The element in the first row and second column of $\exp(At)$ increases without bound as t increases, so there are some initial states $x(0)$ that will result in unbounded $x(t)$. However, there are also some initial states $x(0)$ that will result in bounded $x(t)$. For example, if $x(0) = [1 \ 0 \ 1]^T$, then

$$\begin{aligned} x(t) &= \begin{bmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ e^{-t} \end{bmatrix} \end{aligned} \quad (1.125)$$

and $x(t)$ will be bounded for all t . However, this does not say anything about the stability of the system; it only says that there exists some $x(0)$ that results in a bounded $x(t)$. If we instead choose $x(0) = [0 \ 1 \ 0]^T$, then

$$\begin{aligned} x(t) &= \begin{bmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} t \\ 1 \\ 0 \end{bmatrix} \end{aligned} \quad (1.126)$$

and $x(t)$ increases without bound. This proves that the system is asymptotically unstable and marginally unstable.

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■ EXAMPLE 1.7

Consider the system

$$\dot{x} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} x \quad (1.127)$$

The eigenvalues of A are equal to 0, 0, and -1 . We see that the system is asymptotically unstable since some of the eigenvalues are nonnegative. In order to see if the system is marginally stable, we need to compute the geometric multiplicity of the 0 eigenvalue. (This can be done by noticing that A is already in Jordan form, but we will go through the exercise more completely for the sake of illustration.) Solving the equation

$$(\lambda I - A)v = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (1.128)$$

(where $\lambda = 0$) for nonzero vectors v , we see that there are two linearly independent solutions given as

$$v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (1.129)$$

This shows that the geometric multiplicity of the 0 eigenvalue is equal to 2, which means that the system is marginally stable. The solution of this system is

$$\begin{aligned} x(t) &= \exp(At)x(0) \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} x(0) \end{aligned} \quad (1.130)$$

Regardless of $x(0)$, we see that $x(t)$ will always be bounded, which means that the system is marginally stable. Note that $x(t)$ may approach 0 as t increases, depending on the value of $x(0)$. For example, if $x(0) = [0 \ 0 \ -1]^T$, then

$$x(t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -e^{-t} \end{bmatrix} \quad (1.131)$$

and $x(t)$ approaches 0 as t increases. However, this does not say anything about the asymptotic stability of the system; it only says that there exists some $x(0)$ that results in state $x(t)$ that asymptotically approaches 0. If we instead choose $x(0) = [0 \ 1 \ 0]^T$, then

$$x(t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (1.132)$$

and $x(t)$ does not approach 0. This proves that the system is asymptotically unstable.

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1.6.2 Discrete-time systems

Consider the zero-input, linear, discrete-time, time-invariant system

$$\begin{aligned} x_{k+1} &= Fx_k \\ y_k &= Hx_k \end{aligned} \quad (1.133)$$

The definitions of marginal stability (also called Lyapunov stability) and asymptotic stability are analogous to the definitions for continuous-time systems that were given in Section 1.6.1.

Definition 3 *A linear discrete-time, time-invariant system is marginally stable if the state x_k is bounded for all k and for all bounded initial states x_0 .*

Definition 4 *A linear discrete-time, time-invariant system is asymptotically stable if*

$$\lim_{k \rightarrow \infty} x_k = 0 \quad (1.134)$$

for all bounded initial states x_0 .

Marginal stability and asymptotic stability are types of internal stability. This is because they deal with only the state of the system (i.e., the internal condition of the system) and do not consider the output of the system. More specific categories of internal stability (e.g., uniform stability and exponential stability) are given in some books on linear systems.

Since the solution of Equation (1.133) is given as

$$x_k = A^k x_0 \quad (1.135)$$

we can state the following theorems.

Theorem 5 *A linear discrete-time, time-invariant system is marginally stable if and only if*

$$\lim_{k \rightarrow \infty} A^k \leq M < \infty \quad (1.136)$$

for some constant matrix M . This is just a way of saying that the powers of A do not increase without bound.

Theorem 6 *A linear discrete-time, time-invariant system is asymptotically stable if and only if*

$$\lim_{k \rightarrow \infty} A^k = 0 \quad (1.137)$$

Now recall that $A^k = Q\hat{A}^kQ^{-1}$, where Q is a constant matrix containing the eigenvectors of A , and $\hat{A}$ is the Jordan form of A . The matrix $\hat{A}^k$ therefore contains terms like λ_i^k , $k\lambda_i^k$, $k^2\lambda_i^k$, and so on, where λ_i is an eigenvalue of A . The boundedness of A^k is therefore related to the eigenvalues of A as stated by the following theorems.

Theorem 7 *A linear discrete-time, time-invariant system is marginally stable if and only if one of the following conditions holds.*

1. *All of the eigenvalues of A have magnitude less than one.*